Quantum control for quantum information transfer mediated by topological edge states

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We study quantum state transfer in one-dimensional topological chains based on the Su-Schrieffer-Heeger (SSH) model, comparing two distinct paradigms: domain wall-mediated transfer and counterdiabatic (CD) driving. First, we numerically reproduce the domain wall protocol of Zurita *et al.* [Quantum **7**, 1043 (2023)], achieving fidelities $f > 0.999$ for symmetric two-domain configurations with $L = 11$ sites. We investigate the effect of asymmetric domain wall placement, demonstrating that the fidelity is fundamentally limited by the exponential sensitivity of the effective couplings to the domain lengths. We further explore shortcut-to-adiabaticity (STA) pulse protocols and find that no pulse optimization can overcome the coupling asymmetry that governs the transfer phase. Second, we present a comprehensive analysis of the hybrid analog-digital protocol of Romero *et al.* [Phys. Rev. Applied **21**, 034033 (2024)], where nonadiabatic edge-state transfer is achieved by supplementing the SSH Hamiltonian with approximate counterdiabatic interactions derived from adiabatic gauge potentials and optimized via variational quantum circuits. We offer a systematic comparison of both approaches, contrasting their physical mechanisms, scalability, experimental requirements, and robustness against disorder. The domain wall approach exploits topological protection within an effective low-energy subspace, while the CD approach directly engineers the time evolution to follow the instantaneous eigenstate, achieving fidelities above 99% for chains up to 15 sites in the nonadiabatic regime.

I. INTRODUCTION

Quantum state transfer—the coherent transport of quantum information between distant nodes of a quantum network—is a fundamental primitive for quantum computing and communication architectures [1–3]. In solid-state platforms such as quantum dot arrays, superconducting circuits, or cold-atom lattices, the transfer must contend with disorder, decoherence, and the difficulty of precisely engineering long-range couplings.

A promising approach exploits the *topological protection* of edge states in one-dimensional lattice models [4, 5]. The Su-Schrieffer-Heeger (SSH) model [6, 7]—a paradigmatic example of a symmetry-protected topological insulator in class BDI [8]—supports exponentially localized zero-energy edge states when the system is in the topological phase. These states are robust against local perturbations that respect the protecting chiral symmetry, making them attractive candidates as quantum information carriers.

Zurita, Creffield, and Platero [9] demonstrated that chains comprising multiple SSH domains separated by *topological domain walls* can mediate fast, high-fidelity quantum transfer. The key mechanism is an exponential speed-up arising from the multi-domain structure: by distributing N domain walls along the chain, the transfer time scales as $(w/v)^{\ell/2}$ per domain (where ℓ is the domain length and v/w the hopping ratio) rather than $(w/v)^{L/2}$ for the entire chain. This exponential compression of the transfer time, combined with the topological robustness of the boundary states, represents an efficient and robust approach for quantum information transport.

In this work, we investigate topological quantum state transfer in SSH chains. We focus on two-domain ($N = 2$) configurations and address three questions:

1. Can the transfer protocol of Ref. [9] be faithfully reproduced numerically, and what are the optimal parameters?
2. How does the transfer fidelity degrade when the domain wall is displaced from the chain center, breaking the symmetry between the two domains?
3. Can shortcut-to-adiabaticity (STA) pulse protocols [10–12] improve the transfer performance, particularly in asymmetric configurations?

Additionally, we provide a comprehensive analysis of an alternative approach: the hybrid analog-digital protocol recently proposed by Romero *et al.* [13], where nonadiabatic transfer is achieved through counterdiabatic (CD) driving rather than topological domain walls. We compare both paradigms in terms of their physical mechanisms, scalability, and robustness.

The remainder of this document is organized as follows. Section II introduces the SSH model, its topological properties, and the domain wall construction. Section III describes the transfer protocol and presents our reproduction of the main results of Ref. [9]. Section IV analyzes the effect of asymmetric domain wall placement. Section V investigates STA pulse protocols. Section VI presents a detailed exposition of the hybrid analog-digital counterdiabatic protocol. Section VII provides a systematic comparison of the domain wall and counterdiabatic approaches. Section VIII summarizes our findings and outlines future directions.

II. THE SSH MODEL AND DOMAIN WALLS

A. The Su-Schrieffer-Heeger Hamiltonian

The SSH model describes a one-dimensional tight-binding chain with alternating hopping amplitudes v (intracell) and w (intercell) [6, 7]. For a chain of L sites labeled $j = 0, 1, \ldots, L-1$, the Hamiltonian reads

$$\mathcal{H} = -\sum_{j=0}^{L-2} t_j \big(|j\rangle \langle j+1| + |j+1\rangle \langle j| \big), \tag{1}$$

where

$$t_j = \begin{cases} v & \text{if } j \text{ is even (intracell),} \\ w & \text{if } j \text{ is odd (intercell).} \end{cases} \tag{2}$$

The unit cell contains two sites—sublattice a (even sites) and sublattice b (odd sites). In the thermodynamic limit ($L \to \infty$), the Bloch Hamiltonian in momentum space is $\mathcal{H}(k) = \boldsymbol{d}(k) \cdot \boldsymbol{\sigma}$, where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y)$ and the winding of $\boldsymbol{d}(k)$ around the origin classifies the topological phases.

B. Energy spectrum and topological transition

The bulk dispersion relation is

$$E(k) = \pm\sqrt{v^2 + w^2 + 2vw\cos k}, \tag{3}$$

yielding a spectral gap

$$\Delta = 2|w - v| \tag{4}$$

that closes at $v = w$, signaling the topological phase transition. For $v < w$ the system is in the *topological phase* with winding number $\nu = 1$, while for $v > w$ it is in the *trivial phase* with $\nu = 0$ [5, 14].

Figure 1 shows the energy spectrum of a finite SSH chain ($L = 10$) as a function of v/w. The gap closes linearly at the critical point $v = w$. For a chain with an odd number of sites ($L = 11$), a zero-energy edge state appears in the topological phase, manifesting the bulk-boundary correspondence [15].

C. Topological invariant

The topological classification of the SSH model is captured by the *winding number*

$$\nu = \frac{1}{2\pi} \oint dk \, \frac{d}{dk} \arg\big[d_x(k) + i\, d_y(k)\big], \tag{5}$$

where $d_x(k) = v + w\cos k$ and $d_y(k) = w \sin k$. The winding number counts how many times the vector $\boldsymbol{d}(k)$ encircles the origin as k traverses the Brillouin zone. For the SSH model:

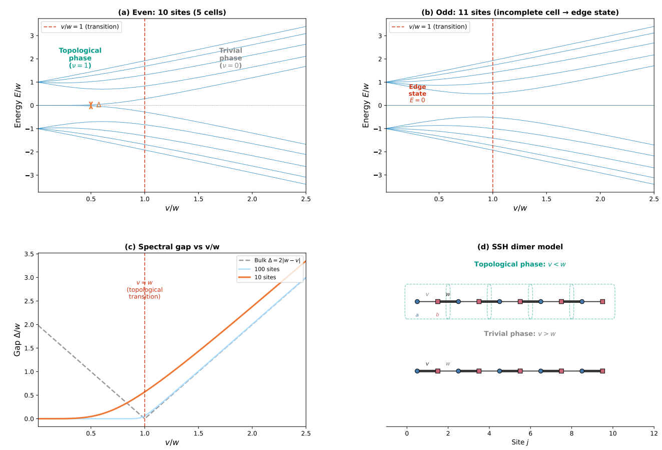


FIG. 1. Energy spectrum of the SSH dimer chain as a function of the hopping ratio v/w. **Left:** Even chain ($L = 10$). The bulk gap $\Delta = 2|w - v|$ closes at $v = w$. **Right:** Odd chain ($L = 11$). A zero-energy state (red) appears in the topological phase ($v/w < 1$), localized at the boundary.

- $v < w$: $\boldsymbol{d}(k)$ encircles the origin $\Rightarrow \nu = 1$ (topological),
- $v > w$: $\boldsymbol{d}(k)$ does not encircle the origin $\Rightarrow \nu = 0$ (trivial).

The winding number is quantized and can only change when the gap closes ($v = w$), reflecting its topological nature. By the bulk-boundary correspondence, a chain with $\nu = 1$ hosts one protected zero-energy state per boundary.

D. Domain walls and protected states

A *domain wall* (DW) is created at the interface between two SSH domains with different dimerization patterns. Consider a chain of total length L composed of two domains meeting at site j_{DW}:

- **Domain 1** ($j < j_{\mathrm{DW}}$): standard SSH hopping pattern $(v, w, v, w, \ldots)$,
- **Domain 2** ($j \geq j_{\mathrm{DW}}$): the pattern restarts as $(v, w, v, w, \ldots)$.

When j_{DW} is at an *odd* site, the bond at the interface involves two consecutive v-bonds (a vv defect), which constitutes a topological domain wall. This is because the dimerization pattern on each side of the interface is conjugate: if the left domain is in the topological phase, the right domain effectively presents the reversed pattern.

For a chain with $N = 2$ domains (one domain wall), the system hosts three topologically protected states near

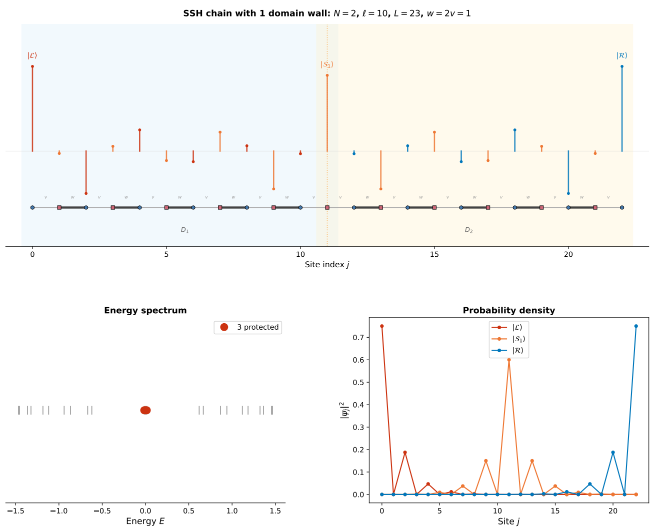


FIG. 2. Topologically protected states in a two-domain SSH chain ($L = 11$, $N = 2$, $\ell = 4$, $v = 0.5$, $w = 1.0$). The three states $|\mathcal{L}\rangle$, $|\mathcal{S}\rangle$, and $|\mathcal{R}\rangle$ are localized at the left edge, the domain wall, and the right edge, respectively. Sublattice polarization (alternating signs) is clearly visible.

$E = 0$ (Fig. 2):

$$|\mathcal{L}\rangle \propto \sum_{n=0}^{\ell_1/2} (-r)^n |2n\rangle , \tag{6}$$

$$\begin{aligned} |\mathcal{S}\rangle \propto |j_{\rm DW}\rangle &+ \sum_{n=1}^{\ell_1/2} (-r)^n |j_{\rm DW} - 2n\rangle \\ &+ \sum_{n=1}^{\ell_2/2} (-r)^n |j_{\rm DW} + 2n\rangle , \end{aligned} \tag{7}$$

$$|\mathcal{R}\rangle \propto \sum_{n=0}^{\ell_2/2} (-r)^n |L - 1 - 2n\rangle , \tag{8}$$

where $r = v/w$ is the decay ratio, and ℓ_1, ℓ_2 are the lengths (number of bonds) of the left and right domains, respectively. Each state resides on a single sublattice and decays exponentially away from its localization center with characteristic length $\xi = -1/\ln r$.

E. Effective low-energy Hamiltonian

The three protected states span an effective low-energy subspace described by

$$\mathcal{H}_{\rm eff} = J_{\mathcal{LS}} |\mathcal{S}\rangle\langle\mathcal{L}| + J_{\mathcal{SR}} |\mathcal{R}\rangle\langle\mathcal{S}| + \text{H.c.}, \tag{9}$$

which has the structure of a three-site tight-binding chain. The effective couplings $J_{\mathcal{LS}}$ and $J_{\mathcal{SR}}$ depend on the overlap of adjacent protected states through the SSH bulk. From the analytical expressions [Eqs. (6)–(8)], these couplings scale as [9]

$$J \sim v \cdot \mathcal{M}_i \cdot \mathcal{M}_j \cdot \left(\frac{v}{w}\right)^{\ell/2} , \tag{10}$$

where ℓ is the domain length separating the two states, and the normalization constants are

$$\mathcal{M}_{\mathcal{L}} = \mathcal{M}_{\mathcal{R}} = \sqrt{\frac{w^2}{v^2} - 1}, \quad \mathcal{M}_{\mathcal{S}} = \sqrt{\frac{w^2 - v^2}{w^2 + v^2}}. \tag{11}$$

Crucially, the exponential dependence of J on ℓ implies that even a small asymmetry $\ell_1 \neq \ell_2$ leads to vastly different effective couplings:

$$\frac{J_{\mathcal{LS}}}{J_{\mathcal{SR}}} \sim \left(\frac{w}{v}\right)^{(\ell_2 - \ell_1)/2} . \tag{12}$$

For our parameters $w = 2v$, a difference $|\ell_1 - \ell_2| = 2$ already produces a coupling ratio of 4, while $|\ell_1 - \ell_2| = 8$ yields a ratio of $2^4 = 16$.

III. TRANSFER PROTOCOL

A. Adiabatic preparation and readout

The transfer protocol proposed by Zurita *et al.* [9] consists of three phases:

1. **Preparation** ($0 \leq t < t_{\rm prep}$): The intracell hopping is ramped from $v = 0$ (fully dimerized) to $v = v_{\rm tr}$ following a smooth pulse shape. This adiabatically maps the initial localized state $|0\rangle$ onto the protected boundary state $|\mathcal{L}\rangle$.

2. **Transfer** ($t_{\rm prep} \leq t < t_{\rm tr} - t_{\rm prep}$): The hopping is held constant at $v = v_{\rm tr}$. The effective couplings $J_{\mathcal{LS}}$, $J_{\mathcal{SR}}$ drive Rabi-like oscillations in the protected subspace, transferring population from $|\mathcal{L}\rangle$ to $|\mathcal{R}\rangle$ via $|\mathcal{S}\rangle$.

3. **Readout** ($t_{\rm tr} - t_{\rm prep} \leq t \leq t_{\rm tr}$): The hopping is ramped back to $v = 0$, adiabatically mapping $|\mathcal{R}\rangle$ back onto the localized state $|L - 1\rangle$.

The standard pulse shape (Eq. (11) of Ref. [9]) is

$$v(t) = \begin{cases} v_{\rm tr} \sin^2(\Omega t) & 0 \leq t < t_{\rm prep}, \\ v_{\rm tr} & t_{\rm prep} \leq t < t_{\rm tr} - t_{\rm prep}, \\ v_{\rm tr} \sin^2(\Omega(t - t_{\rm tr})) & t_{\rm tr} - t_{\rm prep} \leq t \leq t_{\rm tr}, \end{cases} \tag{13}$$

with $\Omega = \pi/(2t_{\rm prep})$. The $\sin^2$ ramp is chosen for its smoothness: both $v(t)$ and $\dot{v}(t)$ vanish at $t = 0$ and $t = t_{\rm prep}$, minimizing diabatic transitions to bulk states. The adiabatic condition requires $t_{\rm prep} \gg \tau = 2/\Delta$, where Δ is the instantaneous spectral gap.

The transfer fidelity is defined as the occupation probability of the target site at the end of the protocol:

$$f = \left|\langle L - 1|\psi(t_{\rm tr})\rangle\right|^2. \tag{14}$$

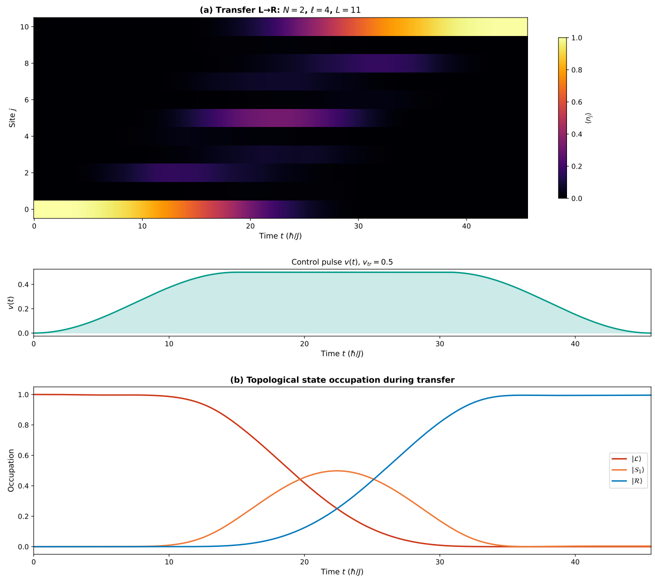


FIG. 3. Transfer protocol dynamics for the symmetric two-domain SSH chain ($L = 11$, $\ell = 4$, $v_{\mathrm{tr}} = 0.5$, $w = 1.0$, $t_{\mathrm{prep}} = 15$, $t_{\mathrm{tr}} = 45$). Population is transferred from site 0 to site 10 via the topologically protected subspace, with final fidelity $f = 0.999$.

B. Transfer time

For a symmetric two-domain chain ($\ell_1 = \ell_2 = \ell$), the effective Hamiltonian (9) has equal couplings $J_{\mathcal{LS}} = J_{\mathcal{SR}} \equiv J$. The time for complete population transfer from $|\mathcal{L}\rangle$ to $|\mathcal{R}\rangle$ via the intermediate state $|\mathcal{S}\rangle$ in a uniform three-site chain is

$$t_{\mathrm{tr}}^{\mathrm{eff}} = \frac{\pi}{\sqrt{2}\,J}, \tag{15}$$

which, using Eq. (10), scales exponentially with the domain length:

$$t_{\mathrm{tr}} \sim \frac{\pi(w^2+v^2)}{2(w^2-v^2)} \left(\frac{w}{v}\right)^{\ell/2+1}. \tag{16}$$

C. Numerical results: reproducing Ref. [9]

We simulate the full time-dependent Schrödinger equation for a chain of $L = 11$ sites ($N = 2$ domains, $\ell = 4$) with $w = 1.0$, $v_{\mathrm{tr}} = 0.5$, and $t_{\mathrm{prep}} = 15$. The time evolution is performed via exact exponentiation of the Hamiltonian at each time step $\Delta t = 0.5$.

Figure 3 shows the transfer dynamics for $t_{\mathrm{tr}} = 45.0$. The initial state $|0\rangle$ evolves through the three protected states: population flows from the left edge to the domain wall and then to the right edge, achieving a final fidelity $f = 0.999$.

Figure 4 presents the fidelity as a function of t_{tr}, showing Rabi-like oscillations with the first maximum at $t_{\mathrm{tr}} \approx 45$, in excellent agreement with the theoretical prediction $t_{\mathrm{tr}} \approx 45.6$ from Eq. (16) and the value reported in Ref. [9]. The peak fidelity $f = 0.999$ exceeds the threshold $f_0 = 0.995$ commonly cited for fault-tolerant quantum error correction [16].

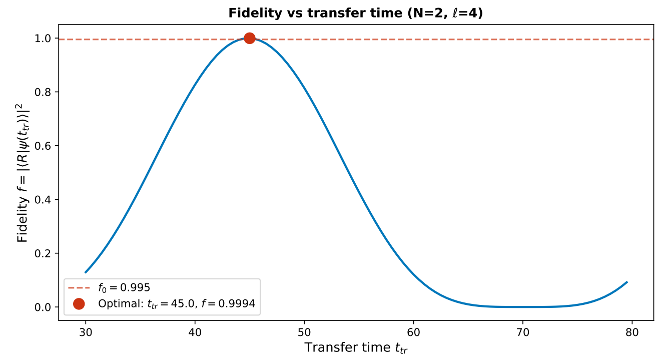


FIG. 4. Transfer fidelity as a function of the total protocol time t_{tr} for the symmetric chain ($L = 11$, $\ell = 4$). The first maximum occurs at $t_{\mathrm{tr}} \approx 45$ with $f = 0.999$, consistent with Ref. [9]. Subsequent oscillations arise from the Rabi dynamics of the effective three-site chain.

IV. ASYMMETRIC DOMAIN WALL PLACEMENT

A. Motivation

In practice, fabrication imperfections may shift the domain wall away from the ideal center of the chain. Furthermore, for chains of certain lengths, it is geometrically impossible to place the domain wall exactly at the center. The domain wall position j_{DW} must be at an *odd-index* site for the vv-bond defect to occur; thus, only chains with $L = 4k + 3$ (i.e., $L = 11, 15, 19, 23, \ldots$) admit a perfectly centered domain wall with $\ell_1 = \ell_2$.

We study a chain of $L = 21$ sites (10 complete unit cells plus one boundary site), varying the domain wall position among three representative odd sites:

- $j_{\mathrm{DW}} = 11$ (quasi-center): $\ell_1 = 11$, $\ell_2 = 9$,
- $j_{\mathrm{DW}} = 7$ (one-third): $\ell_1 = 7$, $\ell_2 = 13$,
- $j_{\mathrm{DW}} = 5$ (one-quarter): $\ell_1 = 5$, $\ell_2 = 15$.

B. Protected states in asymmetric configurations

Regardless of the domain wall position, the system always hosts three protected states near $E = 0$, given by Eqs. (6)–(8) with $\ell_1 \neq \ell_2$. We verify this numerically by diagonalizing the full Hamiltonian and comparing the three eigenstates closest to $E = 0$ with the analytical predictions. Table I summarizes the results.

FIG. 5. Spatial profiles of the three protected states for different domain wall positions in an $L = 21$ chain. Top to bottom: $j_{\rm DW} = 11$ (quasi-center), $j_{\rm DW} = 7$ (one-third), $j_{\rm DW} = 5$ (one-quarter). The exponential localization and sublattice polarization are preserved in all cases.

Figure 5 illustrates how the spatial profiles of the protected states change with the domain wall position. As the wall moves toward the left edge, $|\mathcal{L}\rangle$ becomes increasingly compressed into a shorter domain (fewer sites available for its exponential tail), while $|\mathcal{R}\rangle$ extends over a longer domain. The state $|\mathcal{S}\rangle$ develops an asymmetric envelope, decaying faster into the shorter domain.

TABLE I. Protected states for the asymmetric domain wall configurations in an $L = 21$ chain ($v = 0.5$, $w = 1.0$). The subspace overlap is computed as the singular values of the overlap matrix between analytical and numerical eigenstates.

$j_{\rm DW}$	$\ell_1 + \ell_2$	$E_\pm$	Min. overlap	$J_{\mathcal{LS}}/J_{\mathcal{SR}}$
11	$11 + 9$	± 0.024	0.9998	0.50
7	$7 + 13$	± 0.043	0.9994	8.0
5	$5 + 15$	± 0.086	0.9978	32.0

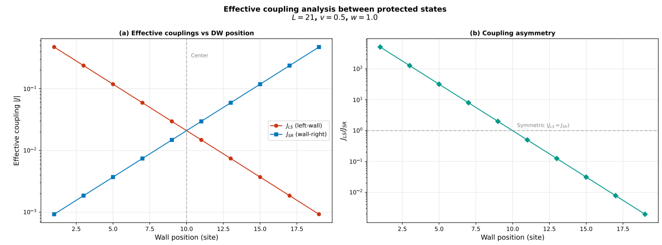


FIG. 6. Effective couplings $J_{\mathcal{LS}}$ (left–soliton) and $J_{\mathcal{SR}}$ (soliton–right) as a function of the domain wall position for $L = 21$. The coupling asymmetry grows exponentially with the displacement from the chain center, consistent with Eq. (12).

C. Effective couplings and the asymmetry mechanism

The effective couplings $J_{\mathcal{LS}}$ and $J_{\mathcal{SR}}$ are computed numerically by projecting the full Hamiltonian onto the protected subspace. The results (Fig. 6) confirm the exponential scaling predicted by Eq. (12):

The coupling asymmetry has a direct and significant consequence for the transfer fidelity. In a three-site chain with unequal hoppings $J_1 \neq J_2$, the maximum achievable population transfer from site 1 to site 3 is bounded by [2]

$$f_{\max} = \frac{4J_1^2 J_2^2}{(J_1^2 + J_2^2)^2}, \tag{17}$$

which equals unity only when $J_1 = J_2$ and decreases rapidly with the coupling ratio. For our quasi-centered case ($J_{\mathcal{LS}}/J_{\mathcal{SR}} = 0.5$): $f_{\max} = 4 \times 0.25/(1.25)^2 = 16/25 = 0.64$.

D. Transfer fidelity in asymmetric chains

Figure 7 shows the transfer dynamics for the three asymmetric configurations, using the standard $\sin^2$ pulse with $t_{\rm prep} = 15$. Table III reports the optimal fidelities obtained from a scan over $t_{\rm tr} \in [30, 600]$.

The quasi-centered case ($j_{\rm DW} = 11$) achieves $f = 0.637$, in quantitative agreement with the analytical bound $f_{\max} = 0.64$ from Eq. (17). The small discrepancy arises from residual diabatic excitations during the ramp phases. For $j_{\rm DW} = 7$ and 5, the fidelity drops to $f = 0.061$ and $f = 0.004$, respectively—effectively rendering the transfer useless.

TABLE II. Effective couplings for the three domain wall positions ($L = 21$, $v = 0.5$, $w = 1.0$).

$j_{\rm DW}$	$J_{\mathcal{LS}}$	$J_{\mathcal{SR}}$	Ratio	Predicted ratio
11	0.0148	0.0296	0.50	$(v/w)^1 = 0.50$
7	0.0593	0.0074	8.0	$(w/v)^3 = 8.0$
5	0.1186	0.0037	32.0	$(w/v)^5 = 32.0$

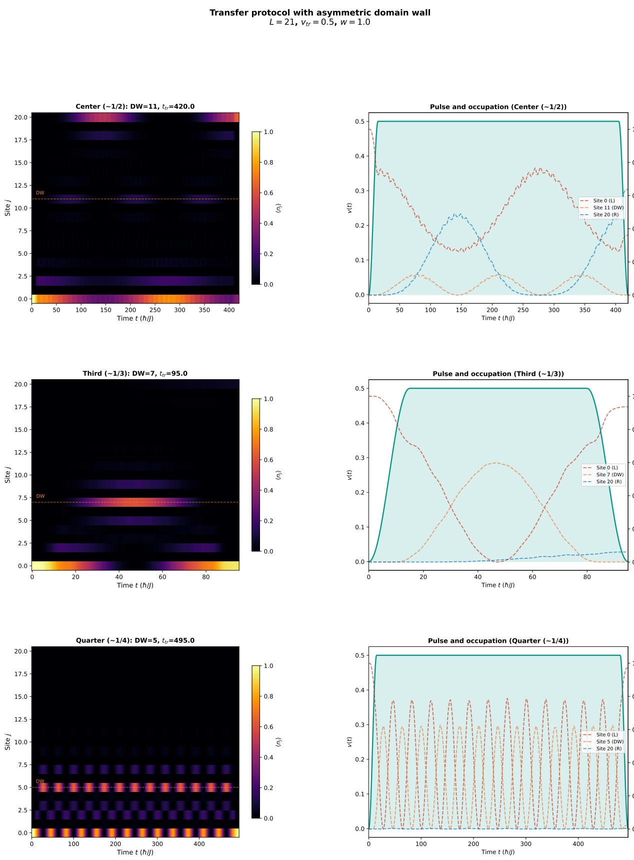


FIG. 7. Transfer dynamics for asymmetric domain wall configurations ($L = 21$). The fidelity degrades dramatically as the domain wall moves away from the chain center.

Figure 8 presents a direct comparison of the fidelity-vs-t_{tr} curves for all three wall positions. The oscillation frequencies and amplitudes differ markedly, reflecting the distinct effective coupling landscapes.

E. Systematic study: fidelity vs. wall position

To map out the full dependence of the transfer fidelity on domain wall position, we perform systematic scans for chains of length $L = 11$ and $L = 15$, placing the domain wall at all allowed (odd) sites. Figure 9 shows that the fidelity profile is perfectly symmetric about the chain center (reflecting the equivalence of swapping the two domains) and sharply peaked:

TABLE III. Optimal transfer fidelity for asymmetric domain wall positions ($L = 21$, standard $\sin^2$ pulse).

j_{DW}	$\ell_1 + \ell_2$	$t_{\mathrm{tr}}^{\mathrm{opt}}$	$f_{\max}$
11 (quasi-center)	$11 + 9$	420	0.637
7 (one-third)	$7 + 13$	95	0.061
5 (one-quarter)	$5 + 15$	495	0.004

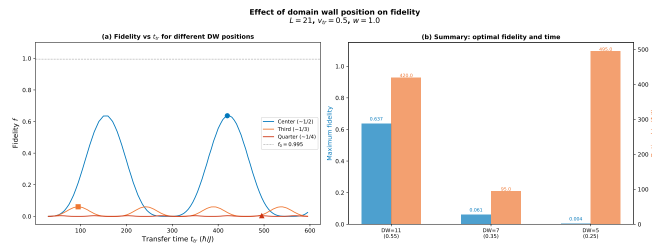


FIG. 8. Transfer fidelity as a function of t_{tr} for the three domain wall positions in the $L = 21$ chain. The first fidelity maximum (marked by dots) decreases exponentially with the displacement from the center.

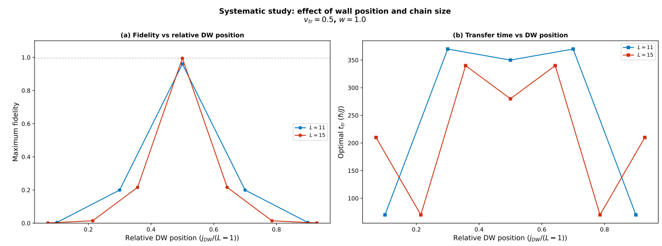


FIG. 9. Maximum transfer fidelity as a function of domain wall position for $L = 11$ (left) and $L = 15$ (right). The fidelity is sharply peaked at the chain center and symmetric about it. Only odd-index sites produce valid domain walls.

- $L = 11$: peak fidelity $f = 0.96$ at $j_{\mathrm{DW}} = 5$ (center), dropping to $f = 0.20$ at $j_{\mathrm{DW}} = 3$ and $f = 0.004$ at $j_{\mathrm{DW}} = 1$.
- $L = 15$: peak fidelity $f = 0.994$ at $j_{\mathrm{DW}} = 7$ (center), dropping to $f = 0.22$ at $j_{\mathrm{DW}} = 5$ and $f = 0.014$ at $j_{\mathrm{DW}} = 3$.

The symmetric profiles confirm that the transfer fidelity depends only on the *difference* $|\ell_1 - \ell_2|$ between domain lengths, not on their individual values. The exponential sensitivity to this difference—inherited from Eq. (12)—imposes stringent requirements on the precision of the domain wall placement in experimental implementations.

V. SHORTCUT-TO-ADIABATICITY PROTOCOLS

A. Motivation and theoretical framework

The preparation and readout phases of the transfer protocol involve adiabatic ramps of the intracell hopping $v(t)$. When the ramp time t_{prep} is not sufficiently long compared to the inverse gap $\tau = 2/\Delta$, diabatic excitations to bulk states reduce the fidelity. Shortcut-to-adiabaticity (STA) techniques [10–12, 17] aim to suppress these excitations by modifying the pulse shape, potentially allowing faster operation without sacrificing fidelity.

The exact counter-diabatic Hamiltonian [11]

$$H_{\mathrm{CD}}(t) = i\hbar \sum_n \left(|\dot{n}(t)\rangle\langle n(t)| - \langle n(t)|\dot{n}(t)\rangle \, |n(t)\rangle\langle n(t)| \right) \tag{18}$$

cancels all diabatic transitions exactly but requires non-local interactions that are impractical to implement. Instead, we explore *approximate* STA strategies through pulse shape optimization.

B. Pulse profiles

We compare four pulse profiles for the ramp phases (Fig. 10):

1. **Standard** $\sin^2$ [Eq. (13)]: The reference protocol from Ref. [9].
2. **STA-α**: A modified ramp $v(t) = v_{\mathrm{tr}} \sin^{2\alpha}(\Omega t)$ with $\alpha > 1$. The higher exponent produces a steeper ramp that spends less time in the intermediate-v region where the gap is smaller. We test $\alpha = 2$ and $\alpha = 3$.
3. **STA-global** $\sin^2$: A single smooth profile over the entire protocol:

$$v(t) = v_{\mathrm{tr}} \sin^2\left(\frac{\pi t}{t_{\mathrm{tr}}}\right), \tag{19}$$

   which has no plateau phase and is infinitely differentiable everywhere.
4. **Linear ramp**: $v(t) = v_{\mathrm{tr}} \min(t/t_{\mathrm{prep}}, 1, (t_{\mathrm{tr}} - t)/t_{\mathrm{prep}})$, serving as a baseline with discontinuous $\dot{v}$ that generates stronger diabatic excitations.

C. Results: symmetric chain

Table IV and Fig. 10 summarize the performance of each pulse profile on the symmetric chain ($L = 11$, $\ell = 4$). The key findings are:

- The standard $\sin^2$ pulse with $t_{\mathrm{prep}} = 15$ provides the optimal balance between speed and fidelity, achieving $f = 0.9994$ at $t_{\mathrm{tr}} = 45$—reproducing the result of Ref. [9].
- The STA-global $\sin^2$ pulse achieves the highest absolute fidelity ($f = 0.99995$) but at the cost of a 67% longer protocol time ($t_{\mathrm{tr}} = 75$ vs. 45).
- The STA-α pulses with reduced $t_{\mathrm{prep}} = 10$ *do not* improve upon the standard protocol. The steeper ramp increases diabatic excitations rather than suppressing them, as the system spends insufficient time adjusting to the changing Hamiltonian.
- The linear ramp performs worst in terms of speed ($t_{\mathrm{tr}} = 97$) due to the discontinuities in $\dot{v}$ at the ramp endpoints, which generate diabatic excitations.

TABLE IV. Pulse comparison for the symmetric chain ($L = 11$, $\ell = 4$, $v_{\mathrm{tr}} = 0.5$, $w = 1.0$).

Pulse	t_{prep}	$t_{\mathrm{tr}}^{\mathrm{opt}}$	$f_{\max}$
Standard $\sin^2$	15	45	0.9994
Standard $\sin^2$	10	89	0.9928
Standard $\sin^2$	5	33	0.9290
STA $\alpha = 2$	10	41	0.9706
STA $\alpha = 3$	10	43	0.9398
STA-global $\sin^2$	–	75	0.99995
Linear	15	97	0.9924

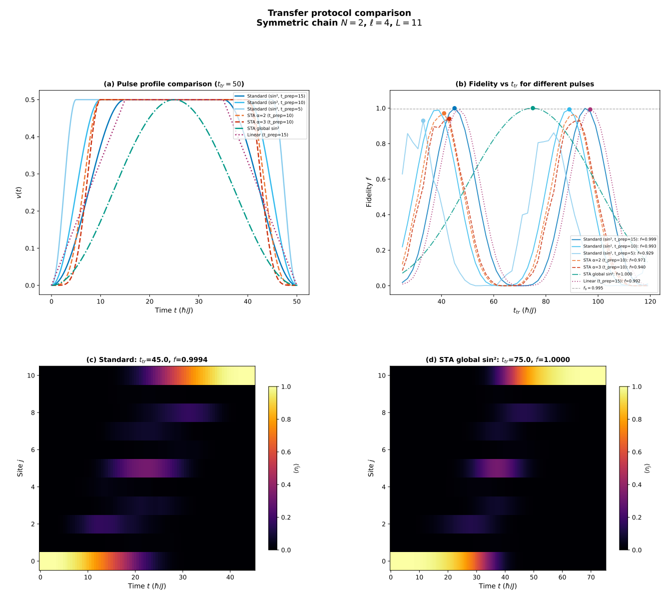


FIG. 10. Comparison of transfer fidelity vs. t_{tr} for different pulse profiles in the symmetric chain ($L = 11$, $\ell = 4$). The standard $\sin^2$ pulse with $t_{\mathrm{prep}} = 15$ achieves the best time–fidelity trade-off ($f = 0.9994$ at $t_{\mathrm{tr}} = 45$). The STA-global $\sin^2$ achieves the highest absolute fidelity ($f = 0.99995$) but requires $t_{\mathrm{tr}} = 75$.

D. Preparation time dependence

Figure 11 shows the dependence of the optimal fidelity on the preparation time t_{prep} for the standard and STA-α pulses. The fidelity follows a characteristic sigmoid: for $t_{\mathrm{prep}} \ll \tau$ the ramp is too fast and diabatic transitions dominate, while for $t_{\mathrm{prep}} \gtrsim \tau$ the fidelity saturates near its maximum. The crossover occurs at $t_{\mathrm{prep}} \sim 2/\Delta_{\min}$, where $\Delta_{\min}$ is the minimum spectral gap encountered during the ramp.

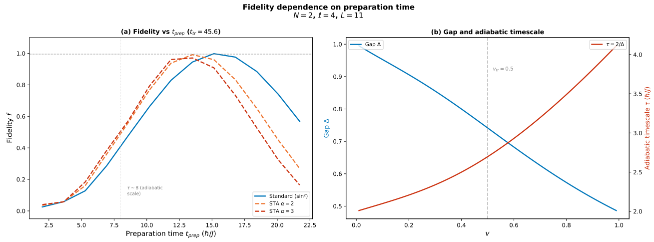
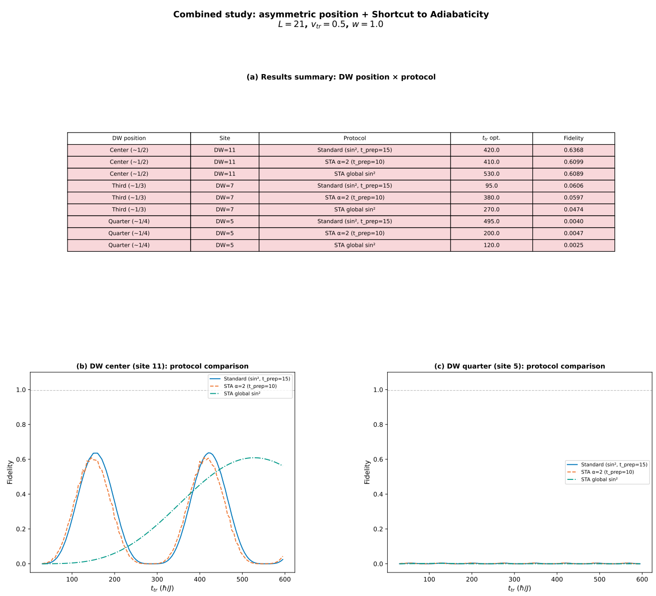


FIG. 11. Optimal transfer fidelity as a function of preparation time t_{prep} for different pulse profiles. The fidelity saturates once t_{prep} exceeds the adiabatic timescale $\tau \sim 2/\Delta_{\min}$.



DW position	Site	Protocol	t_{tr} opt.	Fidelity
Center (~1/2)	DW=11	Standard (sin², t_prep=15)	420.0	0.6368
Center (~1/2)	DW=11	STA α=2 (t_prep=10)	410.0	0.6099
Center (~1/2)	DW=11	STA global sin²	530.0	0.6089
Third (~1/3)	DW=7	Standard (sin², t_prep=15)	95.0	0.0606
Third (~1/3)	DW=7	STA α=2 (t_prep=10)	380.0	0.0597
Third (~1/3)	DW=7	STA global sin²	270.0	0.0474
Quarter (~1/4)	DW=5	Standard (sin², t_prep=15)	495.0	0.0040
Quarter (~1/4)	DW=5	STA α=2 (t_prep=10)	200.0	0.0047
Quarter (~1/4)	DW=5	STA global sin²	120.0	0.0025

FIG. 12. Combined effect of asymmetry and STA pulses on the transfer fidelity ($L = 21$). No pulse optimization can overcome the fundamental coupling asymmetry that limits the maximum achievable fidelity.

E. Results: asymmetric chains with STA

A natural question is whether STA protocols can compensate for the fidelity degradation caused by asymmetric domain wall placement. We test the standard, STA-$\alpha = 2$, and STA-global $\sin^2$ pulses on the three asymmetric configurations ($L = 21$, $j_{\mathrm{DW}} = 11, 7, 5$).

The results (Table V and Fig. 12) are unambiguous: *no pulse protocol significantly improves the fidelity in asymmetric configurations*. This confirms that the limitation is not due to diabatic excitations during the ramp phases—which STA protocols are designed to address—but rather stems from the fundamental coupling asymmetry $J_{\mathcal{LS}} \neq J_{\mathcal{SR}}$ in the transfer phase. The pulse shape affects only the preparation and readout fidelity; the transfer dynamics are governed entirely by the topological properties of the chain.

TABLE V. Pulse comparison for asymmetric configurations ($L = 21$). No pulse protocol significantly improves the fidelity, confirming that the limitation is topological.

j_{DW}	Standard	STA $\alpha = 2$	STA-global
11 (quasi-center)	0.637	0.610	0.609
7 (one-third)	0.061	0.060	0.047
5 (one-quarter)	0.004	0.005	0.003

VI. HYBRID ANALOG-DIGITAL COUNTERDIABATIC PROTOCOL

In this section, we present a detailed analysis of the alternative transfer paradigm proposed by Romero, Chen, Platero, and Ban [13]. Rather than exploiting topological domain walls as in the Zurita protocol (Sec. III), this approach achieves nonadiabatic edge-state transfer in a *uniform* SSH chain by supplementing the Hamiltonian with approximate counterdiabatic (CD) interactions. The CD terms are derived analytically from adiabatic gauge potentials and can be further optimized digitally via variational quantum circuits, hence the denomination "hybrid analog-digital."

A. SSH chain with an odd number of sites

The starting point is a standard SSH chain with $2N-1$ sites (an odd number), with N sites on sublattice A (odd indices) and $N-1$ on sublattice B (even indices). The Hamiltonian is [13]

$$H_0(t) = t_2(t) \sum_{j=1}^{N-1} c_{2j}^\dagger c_{2j-1} + t_1(t) \sum_{j=1}^{N-1} c_{2j+1}^\dagger c_{2j} + \text{H.c.}, \tag{20}$$

where $c_i^\dagger$ (c_i) creates (annihilates) a fermion at site i, $t_2(t)$ is the intra-cell hopping, and $t_1(t)$ is the inter-cell hopping. Due to chiral symmetry, the spectrum consists of $2N-2$ nonzero-energy levels arranged symmetrically about $E = 0$, plus one exact zero-energy eigenstate [5]:

$$|\Phi_0(t)\rangle = \frac{1}{\mathcal{N}} \sum_{i=1}^{N} \left[-\frac{t_2(t)}{t_1(t)} \right]^{i-1} |A_i\rangle, \tag{21}$$

where $\mathcal{N}^2 = \sum_{i=0}^{N-1} [t_2(t)/t_1(t)]^{2i}$ is the normalization factor. Crucially, this zero-energy state has nonzero components only on sublattice A sites (the $b_i = 0$ property), a consequence of chiral symmetry that will be exploited in the simplification of the CD terms.

The transfer mechanism is conceptually simple: when $t_1(0)/t_2(0) \gg 1$, the zero-energy state is localized at the left edge (site A_1). When $t_1(T)/t_2(T) \ll 1$ at the final time $t = T$, the state migrates to the right edge (site A_N). An adiabatic modulation of the hopping parameters achieves this transfer, as in the original Thouless pumping [18, 19]. A standard parametrization is [13]

$$t_{1,2}(t) = t_0(1 \pm \cos \Omega t), \tag{22}$$

where Ω sets the speed of the protocol and the adiabatic regime corresponds to $\Omega = 0.01\, t_0$ (transfer time $T = \pi/\Omega = 100\pi/t_0$). The transfer fidelity is

$$F(t) = |\langle \Phi_0(t)|\Psi(t)\rangle|^2, \tag{23}$$

measuring how closely the evolved state $|\Psi(t)\rangle$ follows the instantaneous zero-energy eigenstate at all times during the protocol.

It is important to note two key differences with the Zurita protocol (Sec. III A): (i) the chain contains no domain wall—it is a uniform SSH chain with a single dimerization pattern; and (ii) the transfer operates by dynamically sweeping the hopping ratio $t_1(t)/t_2(t)$ through the entire topological phase diagram, rather than by populating the protected subspace at a fixed hopping ratio. The adiabatic condition $T \gg \pi/t_0$ ensures that the state tracks the instantaneous eigenstate, but this requires very long protocol times, making the system vulnerable to decoherence.

B. Counterdiabatic driving from adiabatic gauge potentials

The central idea of counterdiabatic (CD) driving [10, 11] is to add an auxiliary Hamiltonian $H_{\mathrm{cd}}(t)$ to the original $H_0(t)$ such that the total evolution under $H(t) = H_0(t) + H_{\mathrm{cd}}(t)$ exactly tracks the instantaneous eigenstate of $H_0(t)$, even when the adiabatic condition is violated. The exact CD term [11] requires full spectral knowledge of $H_0(t)$ at all times and typically involves highly nonlocal interactions, making it impractical.

Romero *et al.* [13] employ the variational approach to approximate CD driving developed by Sels and Polkovnikov [20]. The approximate CD Hamiltonian at order l takes the form

$$H_{\mathrm{cd}}^{(l)} = \dot\lambda\, \mathcal{A}_\lambda^{(l)}, \tag{24}$$

where $\dot\lambda = \dot t_1(t)$ is the rate of change of the control parameter and $\mathcal{A}_\lambda^{(l)}$ is the approximate adiabatic gauge potential, expanded in terms of nested commutators (NCs) of H_0 and $\partial_\lambda H_0$:

$$\mathcal{A}_\lambda^{(l)} = i \sum_{k=1}^{l} \alpha_k^{(l)} \underbrace{[H_0, [H_0, \ldots, [H_0}_{2k-1}, \partial_\lambda H_0]]]. \tag{25}$$

The time-dependent coefficients $\alpha_k^{(l)}(t)$ are determined variationally by minimizing the action $S_l = \mathrm{tr}[G_l^2]$, where $G_l = \partial_\lambda H_0 - i[H_0, \mathcal{A}_\lambda^{(l)}]$. In the limit $l \to \infty$, the expansion converges to the exact gauge potential. For finite l, one obtains an approximate CD driving whose accuracy improves with order.

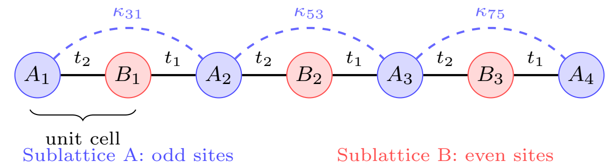


FIG. 13. Schematic of the hybrid analog-digital protocol for a chain of $2N-1 = 7$ sites. The original SSH hopping parameters $t_1(t)$, $t_2(t)$ (solid lines) are supplemented by counterdiabatic NNN hoppings $\kappa_{2j+1,2j-1}$ (dashed arcs) connecting neighboring sublattice A sites. The NNN terms between sublattice B sites are negligible for the zero-energy eigenstate transfer.

C. First-order nested commutator

The first-order ($l = 1$) gauge potential involves only the single commutator $[H_0, \partial_\lambda H_0]$, yielding [13]

$$\mathcal{A}_\lambda^{(1)} = i\kappa^{(1)} \left[\sum_{j=1}^{N-2} c_{2j+2}^\dagger c_{2j} - \sum_{j=1}^{N-1} c_{2j+1}^\dagger c_{2j-1} \right] + \mathrm{H.c.}, \tag{26}$$

where $\kappa^{(1)} = -\Lambda \dot\lambda\, \alpha_1^{(1)}$ with $\Lambda = 2t_0$. The variational parameter is obtained analytically [13, 20] as

$$\alpha_1^{(1)} = -\frac{\mathcal{C}(N)}{t_1^2 + t_2^2}, \quad \mathcal{C}(N) = \frac{2(N-2)+1}{8(N-2)+1}. \tag{27}$$

The physical interpretation of Eq. (26) is highly significant: *the first-order CD driving consists of next-to-nearest-neighbor (NNN) hopping terms between sites of the same sublattice*—odd-odd (sublattice A) and even-even (sublattice B) with equal magnitude $|\kappa^{(1)}|$. Since the zero-energy eigenstate [Eq. (21)] has zero amplitude on all sublattice B sites, the NNN hoppings on sublattice B do not contribute to the transfer and can be neglected. This simplification reduces the CD driving to NNN hoppings exclusively between neighboring sublattice A sites (Fig. 13), which is far more experimentally tractable than the generic nonlocal interactions of the exact CD Hamiltonian (18).

D. Higher-order nested commutators

The first-order NC provides excellent results for short chains (five sites) but becomes insufficient for longer chains, where higher-order corrections are needed. The second-order NC ($l = 2$) introduces additional terms:

$$\mathcal{A}_\lambda^{(2)} = i\big(\alpha_1^{(2)}[H_0, \partial_\lambda H_0] + \alpha_2^{(2)}[H_0, [H_0, [H_0, \partial_\lambda H_0]]]\big), \tag{28}$$

which, after evaluation, produces NNN hoppings (as from the first order) plus fourth-nearest-neighbor hoppings within and between sublattices. Again, the sublattice B terms can be neglected for the zero-energy state transfer.

In general, the lth-order NC introduces interactions up to $2l$th-nearest neighbors [13]. For a chain with $2N-1$ sites, the minimum order required to account for all possible interactions along the same sublattice is $l = N-1$. This rapid growth of computational complexity and long-range interactions motivates two key simplifications performed in Ref. [13]:

1. **Neglect sublattice B hoppings**: Since the transfer trajectory follows the zero-energy eigenstate, which has zero amplitude on sublattice B, the CD hoppings between even-even sites do not contribute.

2. **Retain only NNN hoppings on sublattice A**: The longer-range hoppings (fourth-nearest neighbors and beyond) from higher-order NCs are discarded, keeping only the dominant NNN terms. The simplified CD driving takes the form

$$H_{\rm cd} = -i \sum_{j=1}^{N-1} \kappa_{2j+1,2j-1}\, c_{2j+1}^{\dagger} c_{2j-1} + \text{H.c.}, \qquad (29)$$

where $\kappa_{2j+1,2j-1}$ are site-dependent coupling strengths to be optimized.

This simplified form retains the essential physics of the CD correction while requiring only experimentally accessible NNN hoppings. For a five-site chain, this simplified protocol achieves fidelities indistinguishable from the full second-order NC, confirming that the NNN sublattice A terms capture the dominant CD contribution.

E. Analog counterdiabatic results

For a five-site chain ($N = 3$) in the nonadiabatic regime ($\Omega = t_0$, $T = \pi$), the performance of the different CD approximations can be summarized as follows [13]:

- **No CD** (H_0 only): The fidelity drops significantly in the nonadiabatic regime, falling below 50% for $\Omega \gtrsim 0.1\, t_0$.

- **First-order NC** ($H_0 + H_{\rm cd}^{(1)}$): The transfer improves substantially but does not reach high fidelity for $\Omega = t_0$.

- **Second-order NC** ($H_0 + H_{\rm cd}^{(2)}$): The protocol retains high fidelity $F(T) > 99\%$ even at $\Omega = t_0$ (transfer 100× faster than the adiabatic regime), with CD driving strengths of the same order as t_0.

These results demonstrate that a modest order of the NC expansion suffices for short chains. However, for chains longer than five sites, higher orders become necessary, and the analytical approach becomes computationally expensive. This motivates the variational digital optimization described next.

F. Variational quantum circuit optimization

To circumvent the limitations of the analytical NC approach, Romero *et al.* [13] leverage digital quantum simulation (DQS) to find optimal CD driving parameters. The key idea is to treat the NNN hopping strengths $\kappa_{2j+1,2j-1}$ as variational parameters and optimize them using classical optimizers in a hybrid quantum-classical loop [21].

1. Encoding and Trotterization

The SSH chain of $2N-1$ sites is encoded in $n = \lceil \log_2(2N-1) \rceil$ qubits. Both H_0 and $H_{\rm cd}$ are decomposed in the Pauli basis, and the time evolution is implemented via first-order Trotter-Suzuki decomposition [22]:

$$U(T,0) \approx \prod_{k=1}^{r} e^{-iH_0^{(c)}(k\Delta t)\Delta t}\, e^{-iH_{\rm cd}^{(c)}(k\Delta t)\Delta t}, \qquad (30)$$

with $r = T/\Delta t$ Trotter steps. The NNN coupling strengths are discretized as a vector of $r+1$ parameters per hopping location, with boundary conditions $\kappa(0) = \kappa(T) = 0$, leaving $r-1$ free parameters per location to optimize.

2. Cost functions

Two cost functions are employed [13]:

1. **Transfer fidelity**: Directly maximizing $C(\vec{\kappa}^{(c)}) = F(T)$ [Eq. (23)], which requires computing state overlaps and is well-suited for noiseless simulators.

2. **Hellinger distance**: A measurement-based cost function

$$H^2(P,Q) = 1 - \sum_{i=0}^{m-1} \sqrt{p_i q_i}, \qquad (31)$$

where P is the measured probability distribution after $N_{\rm shot}$ measurements and Q is the target distribution (unit probability at the final site). Minimizing H^2 is equivalent to maximizing the population at the target site and is practical for real quantum hardware [13].

The optimization uses the simultaneous perturbation stochastic approximation (SPSA) algorithm, initialized with values proportional to the analytical first-order NC result $\kappa^{(1)}$ (“warm starting”) and with adaptive bounds to keep $\|H_{\rm cd}\| \sim \|H_0\|$ for experimental feasibility.

3. Two simplification strategies

Two simplification ansatzes are tested for the NNN hoppings:

1. **Time-symmetric hoppings**: Exploiting the symmetry $\kappa^{(c)}_{2i+1,2i-1}(t) = \kappa^{(c)}_{2(N-i)+1,2(N-i)-1}(T-t)$, which halves the number of free parameters to $\lfloor N/2 \rfloor (r-1)$.
2. **Equal NNN hoppings**: All NNN couplings are set equal, $\kappa^{(c)}_{2j+1,2j-1}(t) \mapsto \kappa^{(c)}(t)$, reducing the optimization to only $r-1$ parameters regardless of chain length. This is inspired by the site-independent structure of the first-order NC.

4. Results across chain lengths

The variational circuit is implemented with $r = 22$ Trotter steps, with 10 independent runs per configuration to assess stability. Table VI summarizes the optimal fidelities obtained across chain lengths from 5 to 15 sites [13].

For short chains (5–9 sites), the optimized CD driving achieves near-perfect fidelities with modest optimization effort. For longer chains (11–15 sites), the fidelity decreases to approximately 90% when the constraint $\|H_{\rm cd}\| \sim \|H_0\|$ is maintained. Relaxing this constraint (allowing stronger NNN hoppings) improves the fidelity at the expense of experimental feasibility.

G. Transfer solely by counterdiabatic driving

A remarkable finding of Ref. [13] is that high-fidelity transfer can be achieved by setting $H_0 = 0$ and driving the system entirely through the CD terms. In this regime, the circuit contains only the variational NNN hoppings on sublattice A, which simplifies the implementation in several ways:

- The circuit depth is significantly reduced (no H_0 block),
- Convergence is faster, requiring only 1000 iterations for all chain lengths tested,
- The fidelities consistently exceed 99% for chains up to 15 sites, outperforming the $H_0 + H_{\rm cd}$ scheme.

TABLE VI. Transfer fidelities $F(T)$ from the variational quantum circuit optimization [13] for different chain lengths in the nonadiabatic regime ($\Omega = t_0$, $T = \pi$), using equal NNN hoppings on sublattice A with the transfer fidelity as cost function.

Sites	N	Qubits	Iterations	$\bar{F}(T)$
5	3	3	1000	> 0.999
7	4	3	2000	> 0.99
9	5	4	3000	> 0.99
11	6	4	5000	~ 0.97
13	7	4	6000	~ 0.94
15	8	4	10000	~ 0.90

This counterintuitive result can be understood as follows: when $H_0 = 0$, the population is entirely confined to sublattice A (no excitation of sublattice B sites), and the variational optimizer has full freedom to design the transfer pathway without competition from the original Hamiltonian's dynamics. The optimal CD driving in this case effectively encodes all the information about the SSH topology into a purely sublattice A hopping protocol.

H. Robustness against disorder

The robustness of the CD protocol is assessed against three types of perturbations [13]:

1. **Rice-Mele staggered potential**: A diagonal term $\sum_j (-1)^{j-1}\Delta(t)\, c_j^\dagger c_j$ that breaks chiral symmetry. The fidelity remains above 90% for $\delta/t_0 \lesssim 0.1$.
2. **Diagonal disorder**: Random on-site energies $\delta_j \in [-\sigma_\delta, \sigma_\delta]$. High fidelity is preserved for $\sigma_\delta/t_0 \lesssim 0.1$.
3. **Off-diagonal (hopping) disorder**: Random perturbations to the hopping amplitudes $\tau_j \in [-\sigma_\tau, \sigma_\tau]$. The protocol maintains high fidelity for $\sigma_\tau/t_0 \lesssim 0.1$.

The CD protocol with different (site-dependent) NNN hoppings is generally more robust than the equal-hopping version, as the site-dependent parameters can partially compensate local perturbations. The robustness threshold of $\sim 10\%$ disorder strength relative to t_0 is consistent with the energy scale hierarchy required for the CD correction to remain a perturbative supplement.

VII. COMPARISON OF TRANSFER PARADIGMS

Having analyzed both the domain wall protocol (Secs. III–V) and the hybrid analog-digital CD protocol (Sec. VI), we now provide a systematic comparison of both approaches.

A. Physical mechanism

The two protocols exploit fundamentally different physical mechanisms for edge-state transfer:

Domain wall approach (Zurita *et al.* [9]). The chain contains topological domain walls that create protected boundary states ($|\mathcal{L}\rangle$, $|\mathcal{S}\rangle$, $|\mathcal{R}\rangle$) within the bulk gap. The system is prepared in one boundary state and the hopping amplitude $v(t)$ is ramped adiabatically to populate the target state at the opposite edge. The transfer operates within a *low-energy subspace* spanned by the topologically protected zero-energy states, and the bulk gap provides protection against excitations. The domain wall itself is a static structural feature of the chain.

CD approach (Romero *et al.* [13]). The chain is a uniform SSH model (no domain wall). The hoppings $t_1(t)$, $t_2(t)$ are swept dynamically, and the transfer occurs by following the instantaneous zero-energy eigenstate of $H_0(t)$ through the topological phase diagram. To accelerate this process without losing fidelity, auxiliary NNN hoppings are added as counterdiabatic corrections that suppress nonadiabatic transitions to other eigenstates. The transfer mechanism is a *dynamical shortcut to adiabaticity* rather than a topological subspace approach.

B. Chain structure

A key practical difference concerns the chain topology:

- The **domain wall protocol** requires careful engineering of the chain structure, placing domain walls at specific positions. As demonstrated in Sec. IV, even small deviations from the ideal wall position cause exponential degradation of the transfer fidelity [Eq. (17)]. This imposes stringent fabrication requirements.

- The **CD protocol** operates on a uniform SSH chain with no structural defects. The complexity is shifted from the chain architecture to the control fields: time-dependent NNN hoppings $\kappa(t)$ must be engineered between sublattice A sites. These can be tuned continuously during the protocol.

C. Scalability

Both approaches face scalability challenges, but of different kinds:

Domain wall protocol. The transfer time scales exponentially with the domain length [Eq. (16)]: $t_{\mathrm{tr}} \sim (w/v)^{\ell/2}$. For $w = 2v$ and $\ell = 10$, $t_{\mathrm{tr}} \sim 2^5 = 32$ (in units of $1/v$). The multi-domain construction of Ref. [9] mitigates this by dividing the chain into N shorter domains, reducing the transfer time from $(w/v)^{L/2}$ to $N\times(w/v)^{L/(2N)}$—an exponential improvement. The protocol has been demonstrated numerically for chains up to $L \sim 100$ sites [9], though the number of domain walls must increase proportionally.

CD protocol. The operation time is fixed at $T = \pi/\Omega$ (nonadiabatic regime), which does *not* scale with the chain length. However, the fidelity degrades for longer chains: the approximate CD correction (NNN hoppings only) becomes less accurate as the chain length increases, since higher-order NCs—which introduce increasingly long-range hoppings—are needed but are truncated. With the equal-hopping ansatz and the constraint $\|H_{\mathrm{cd}}\| \sim \|H_0\|$, the fidelity drops to $\sim 90\%$ for a 15-site chain [13]. The $H_0 = 0$ variant maintains $F > 99\%$ for 15 sites but requires engineering a purely CD-driven system without the reference Hamiltonian.

Table VII provides a quantitative comparison.

D. Experimental requirements

The domain wall protocol requires only nearest-neighbor hopping control with a single time-dependent parameter $v(t)$. The domain wall structure is *static* and must be encoded at fabrication time. This makes it naturally suited to platforms with fixed lattice geometry, such as quantum dot arrays [23] or photonic lattices [24].

The CD protocol additionally requires time-dependent NNN hopping terms $\kappa(t)$ between sublattice A sites. While more complex to control, this is feasible in several platforms: in superconducting qubit chains, NNN hoppings can be implemented via geometric phases or artificial gauge fields [13]; in cold atoms, angular momentum states [13] or synthetic dimensions provide tunable long-range couplings. The digital optimization via variational circuits provides discretized solutions that are naturally implementable in gate-based quantum processors.

E. Robustness comparison

The two protocols exhibit different robustness characteristics:

- The domain wall protocol enjoys *topological protection*: the boundary states are robust against any perturbation that respects chiral symmetry. However, perturbations that break chiral symmetry (such as on-site disorder or Rice-Mele potentials) can mix the protected subspace with bulk states. Furthermore, as demonstrated in Sec. IV, the protocol is exquisitely sensitive to the *structural* asymmetry of the domain wall position—a form of "geometric disorder" that no pulse optimization can correct.

- The CD protocol is robust against moderate disorder ($\sigma/t_0 \lesssim 0.1$) without relying on topological protection. The robustness arises from the variational optimization, which learns CD driving profiles that are effective even in the presence of perturbations. However, since the protocol deliberately breaks chiral symmetry by introducing NNN hoppings, it does not benefit from topological protection in the conventional sense.

TABLE VII. Comparison of the two transfer protocols. DW = domain wall, CD = counterdiabatic.

Feature	**DW protocol** [9]	**CD protocol** [13]
Chain structure	Domain walls at specific sites	Uniform SSH chain
Transfer mechanism	Rabi oscillations in topological subspace	Following instantaneous eigenstate via CD
Interactions	Nearest-neighbor only	NN + NNN (sublattice A)
Number of sites	L odd or even, $N \geq 2$ domains	$2N - 1$ sites (odd)
Transfer time	$t_{\mathrm{tr}} \sim (w/v)^{\ell/2}$, exponential	$T = \pi/\Omega$, fixed (nonadiabatic)
Scalability (time)	Exponential; mitigated by multi-domain	Constant time; fidelity decreases with L
Fidelity ($L = 11$, sym.)	$f = 0.999$	$F > 0.97$ (H_0+H_{cd}); F>0.99 (H_0=0)
Sensitivity	Exponential to DW position	No structural sensitivity
Disorder robustness	Topological protection	Robust for $\sigma/t_0 \lesssim 0.1$
Control complexity	Single ramp $v(t)$	Time-dependent NNN $\kappa(t)$ profiles
Optimization	t_{prep}, t_{tr}	$r(N{-}1)$ or r parameters
Experimental platforms	QD, SC qubits, cold atoms	Same; requires NNN coupling tunability

F. Complementarity of the two approaches

The two paradigms are complementary rather than competitors. The domain wall protocol excels in scenarios where:

- The chain structure can be fabricated with high precision,
- Only nearest-neighbor couplings are available,
- Chiral-symmetry-preserving conditions can be maintained,
- Long chains are needed (via multi-domain construction).

The CD protocol is preferable when:

- The chain structure cannot be precisely controlled,
- NNN couplings are experimentally accessible,
- Fast operation (nonadiabatic regime) is required to beat decoherence,
- A uniform chain is simpler to fabricate than one with domain walls.

A natural future direction is to combine both approaches: using domain wall-mediated transfer as the primary mechanism while adding CD corrections to suppress the residual nonadiabatic excitations during the preparation and readout ramps. This "best of both worlds" strategy would leverage topological protection for the transfer phase while using CD driving for the time-critical ramp phases—precisely the regime where the STA protocols of Sec. V were shown to provide only marginal improvements.

VIII. DISCUSSION AND CONCLUSIONS

We have presented a detailed study of quantum state transfer in SSH chains, comparing two distinct paradigms: domain wall-mediated transfer following Zurita *et al.* [9] and hybrid analog-digital counterdiabatic driving following Romero *et al.* [13].

Our main results can be summarized as follows:

1. Faithful reproduction of the topological transfer protocol. For the symmetric configuration ($L = 11$, $N = 2$, $\ell = 4$), we reproduce the results of Ref. [9] with high precision: the optimal transfer time $t_{\mathrm{tr}} = 45$ agrees with the analytical prediction $t_{\mathrm{tr}} \approx 45.6$, and the fidelity $f = 0.999$ exceeds the fault-tolerance threshold.

2. Exponential sensitivity to domain wall position. Moving the domain wall away from the chain center degrades the fidelity according to the fundamental bound $f_{\max} = 4J_1^2 J_2^2/(J_1^2 + J_2^2)^2$. The coupling ratio $J_1/J_2 \sim (w/v)^{|\ell_1 - \ell_2|/2}$ grows exponentially with the displacement, making the fidelity exquisitely sensitive to the wall position. For a single unit-cell displacement in $L = 21$, the fidelity drops from ~ 1 to 0.637; for larger displacements, the transfer becomes negligible.

3. STA protocols cannot overcome topological constraints. Shortcut-to-adiabaticity pulse modifications provide marginal improvements in the symmetric case (the global $\sin^2$ achieves $f = 0.99995$) but are *completely ineffective* against the coupling asymmetry. This establishes a clear separation between the two sources of infidelity: (i) diabatic excitations during the ramp phases, which are addressable by pulse engineering, and (ii) the coupling asymmetry during the transfer phase, which is a topological property of the chain and can only be addressed by modifying the chain structure.

4. Counterdiabatic driving as an alternative paradigm. The hybrid analog-digital protocol of Ref. [13] circumvents the structural sensitivity of the domain wall approach by operating on a uniform SSH chain with time-dependent NNN hoppings as counterdiabatic corrections. The CD terms, derived from adiabatic gauge potentials via nested commutators [20], can be further optimized using variational quantum circuits. In the nonadiabatic regime ($T = \pi/\Omega$, a $100\times$ speed-up over the adiabatic protocol), the CD approach achieves fidelities above 99% for short chains (5–9 sites) and approximately

90% for 15 sites when CD strengths are constrained to the same order as t_0.

5. Complementarity of the two paradigms. The domain wall and CD approaches are complementary: the former excels in platforms with precise structural control and nearest-neighbor-only couplings, while the latter is preferable for uniform chains with tunable NNN interactions and stringent time constraints. A promising future direction is to combine both strategies—using domain wall-mediated transfer as the primary mechanism while adding CD corrections to the preparation and readout phases—leveraging topological protection for the transfer dynamics and counterdiabatic driving for the time-critical ramp phases.

6. Parity constraint on domain wall placement. We identify that domain walls can only be placed at odd-index sites, and that a perfectly centered symmetric configuration requires $L = 4k+3$. This geometric constraint must be accounted for in experimental designs.

These findings provide a comprehensive picture of topological quantum state transfer in SSH chains. The exponential speed-up predicted for multi-domain ($N > 2$) chains [9], the variational optimization strategies for CD driving [13, 25], and the possibility of hybrid domain-wall/CD protocols offer multiple avenues for future investigation and experimental implementation across superconducting qubit, cold-atom, and quantum dot platforms.

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